

Moksha Smruthi Morapakula

Tempe, AZ · Open to Relocate · Open to Remote · Available: June 2026
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EDUCATION

Arizona State University

Tempe, AZ

M.S. Data Science, Analytics & Engineering, Focus: Computing and Decision Analytics

Aug 2024 – May 2026

- **Coursework:** Machine Learning, Deep Learning, Data Mining, Data Visualization, Data Processing at Scale, Statistics for Data Analysts, Data-Driven Decision Making, Information Assurance & Security

SRM University AP

Amaravati, India

B.Tech. Computer Science & Engineering (AI & ML)

Jun 2020 – May 2024

- **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Database Management Systems, Deep Learning, Machine Learning, Computer Networks, Statistics & Probability, Operating Systems

TECHNICAL SKILLS

Programming Languages: Python, SQL (MySQL, PostgreSQL), JavaScript, R, C++, HTML/CSS, PHP, Bash, MATLAB

AI / Machine Learning: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, CNNs, GANs, Fuzzy-Neural Networks, DBSCAN, KMeans, PCA, transfer learning, anomaly detection, causal inference

LLM & AI Integration: Claude API, OpenAI API, Google Gemini, Ollama, multimodal AI, prompt engineering, MCP (Model Context Protocol)

Data Engineering & Pipelines: ETL/ELT pipeline design, multi-source data integration, schema alignment, data validation & quality, large-scale data processing, data warehousing

Analytics & Visualization: Tableau, Power BI, Streamlit, D3.js, Three.js, Plotly, pandas, NumPy, statsmodels, KPI tracking, A/B testing, statistical modeling, time-series forecasting

Platforms & Tools: AWS, Linux, Git/GitHub, Docker (concepts), VS Code, Jupyter, Google Workspace, Microsoft Office Suite, Jira

PROFESSIONAL EXPERIENCE

Research, Data & Content Aide

Oct 2025 – Present

ASU University Design Institute

Tempe, AZ

- Reconciled and analyzed 50,000+ institutional records across 15+ internal and external sources using **SQL** and **Python** (pandas, BeautifulSoup); produced decision-ready outputs for institutional reporting.
- Built and published public-facing charts, dashboards, and visual reports translating complex multi-source datasets into accessible outputs for non-technical higher education audiences; documented data sources, transformation logic, and reporting workflows for repeatable institutional reporting.

Research Intern – Machine Learning & Deep Learning

May 2022 – Dec 2023

SRM University AP

Amaravati, India

- Designed, implemented, and evaluated deep learning architectures including **CNNs**, **Fuzzy Neural Networks**, and **LSTMs** using Python, TensorFlow, and PyTorch across computer vision and sequence classification tasks.
- Built automated data preprocessing pipelines reducing manual effort by **30%** and improving downstream model accuracy from 73% to 85%; developed reliability models and failure prediction patterns to ensure production stability under heavy computational loads.
- Conducted comparative analysis of AI algorithms across hardware environments; applied custom loss functions, transfer learning, and optimization techniques to improve model performance and scalability.
- Produced full technical documentation covering research methodology, implementation procedures, and evaluation metrics, enabling lab-wide reproducibility and seamless research handoff.

Data Systems Engineering Intern

Mar 2024 – Jul 2024

Aganith Pvt Ltd

Hyderabad, India

- Administered and maintained relational databases supporting live client-facing applications; performed data cleaning, transformation, and validation to ensure integrity across multi-source datasets.

- Conducted end-to-end data analysis across client engagements, identifying patterns and anomalies and translating findings into structured reports for consulting team review.
- Collaborated with clients to gather, document, and validate data and system requirements; tracked change requests and ensured accurate implementation across reporting cycles.
- Executed systematic report testing and quality assurance across multiple client deliverables, verifying output accuracy against defined specifications before release.
- Supported cross-functional consulting teams in delivering data-related work on time under client confidentiality and non-disclosure agreements across industries including maritime, finance, and enterprise software.

Data Analysis & Content Reporting Intern

May 2022 – Apr 2024

IR & Higher Studies Office, SRM University AP

Amaravati, India

- Analyzed outreach and engagement datasets from **30+ countries** to identify international partnership trends; translated findings into executive-facing reports that increased stakeholder engagement by **25%**.
- Built and maintained data tracking workflows for cross-institutional outreach programs; cleaned and reconciled records across multiple sources to ensure data accuracy and reporting consistency.
- Led end-to-end publication workflow for the annual International Relations magazine, coordinating cross-functional contributors, managing editorial data, and reducing production time by **30%** through systematic documentation and process improvements.
- Authored research-driven reports on higher education policy and international partnerships; managed sensitive cross-border correspondence and stakeholder communications for university leadership.

TECHNICAL PROJECTS

Causal Impact of Remote Learning on U.S. Student Recovery | *Python, pandas, statsmodels, XGBoost, SHAP, Streamlit*

- Engineered a clean analytical asset by merging 4 independent government sources (Stanford SEDA, CDC, NCES, Census SAIPE) on NCES district ID, resolving encoding inconsistencies, zero-padding mismatches, and subgroup filtering across **44,000+ district-year observations**.
- Designed a Difference-in-Differences model with district and year fixed effects to isolate the causal effect of remote instruction on post-COVID test score recovery; applied HC3 robust standard errors and validated parallel trends across 3 model specifications.
- Trained XGBoost classifier to predict at-risk districts; applied SHAP decomposition to quantify marginal contributions of poverty rate, urbanicity, and broadband access, translating black-box ML into interpretable policy signals.
- Deployed a public-facing Streamlit dashboard with U.S. choropleth maps, interactive trend visualization, and multi-model comparison; full-stack ownership from raw ingestion to production output.

Dynamic Pricing for Perishable Inventory | *Python, XGBoost, Prophet, scikit-learn, SQL, Tableau*

- Designed a five-stage ML pipeline integrating two heterogeneous retail datasets (Dunnhumby Complete Journey + Kaggle perishables, **100,000+ records, 45,301 SKUs**) via category-level schema alignment and ETL.
- Built demand forecasting (XGBoost + Facebook Prophet, 1–7 day horizon), shelf-life risk classification (AUC-ROC ≥ 0.70), log-log price elasticity estimation, grid-search markdown optimization (0–40% discount), and a 30-day simulation comparing pricing strategies.
- Engineered perishable SKU features (days until expiry, spoilage risk, promo flags, lag sales); managed sparse demand via category-level aggregation when SKU transaction counts fell below 30.

GreenTrace – Satellite-Verified Supply Chain Intelligence | *Python, Google Earth Engine, ESA Sentinel-5P, Streamlit*

- Built an end-to-end emissions intelligence platform cross-referencing company-declared shipping emissions against ESA Sentinel-5P NO2 satellite data (TROPOMI L3 via Google Earth Engine APIs) across 7 global corridors.
- Engineered a Discrepancy Index pipeline using GLEC v3.2 / ISO 14083 emission factors; flagged 4/7 corridors with anomalies averaging **+35%** above declared levels; delivered findings via client-facing Streamlit dashboard with composite scoring and interactive route optimization.

Outfit Roaster – Multimodal AI Fashion Analysis | *Voxel51 FiftyOne, Google Gemini – 1st Place, Voxel51 Hackathon*

- Built a multimodal AI application integrating Voxel51 FiftyOne computer vision dataset management with Google Gemini vision-language model to analyze outfit images and generate real-time fashion critiques.
- Designed an image ingestion and metadata tagging pipeline; engineered end-to-end LLM API orchestration from vision input to structured natural-language output under hackathon time constraints.

Customer Segmentation & Market Campaign Optimization | *Python, SQL, Tableau, scikit-learn, PCA*

- Processed and analyzed ~**541,000** e-commerce records; developed RFM pipelines using KMeans and DBSCAN clustering (Silhouette Score: 0.44); applied PCA for high-dimensional visualization and cluster validation.
- Identified segments from VIP to budget-sensitive customers; built interactive Tableau dashboards contributing to a **22% improvement** in campaign conversion rates.

Financial Risk Reporting & Fraud Detection | *Python, SQL, Tableau, scikit-learn*

- Analyzed **6,000+** institutional compliance records; performed Z-score normalization and correlation analysis identifying 5 key risk factors predicting **85%** of fraudulent activity.
- Built Tableau dashboards with drill-down filters, KPI tracking, and time-series heatmaps; documented transformation logic and validation rules for repeatable production reporting.

Skin Cancer Classification with InceptionV3 | *Python, TensorFlow, OpenCV, AWS*

- Built CNN model using InceptionV3 transfer learning achieving **91% classification accuracy** on medical image dataset; deployed large-scale preprocessing pipeline on AWS cloud infrastructure.
- Evaluated model robustness through ROC analysis ensuring high sensitivity and specificity for clinical decision support.

Interactive Space Mission Scrollytelling Website | *HTML, D3.js, Three.js, Python, JavaScript*

- Built a full-stack scrollytelling web application visualizing 60+ years of NASA/UNOOSA space mission data (**4,000+ launch records**) with 10+ custom visualizations, 3D components, and animated SVGs.
- Designed ETL pipeline transforming raw datasets into scalable web-accessible formats; added ARIA attributes and keyboard navigation for full accessibility compliance.

LEADERSHIP & INVOLVEMENT

Volunteer, International Student & Scholars Center

Aug 2024 – Dec 2025

Arizona State University

Tempe, AZ

- Facilitated onboarding and orientation events for **200+ international students**; coordinated student inflow across cultural and community events throughout the academic year.
- Analyzed post-event feedback data to identify improvement opportunities, contributing to a **20% increase** in satisfaction scores.

Vice President

Dec 2021 – Dec 2022

Anime Society, SRM University AP

Amaravati, India

- Led cultural and community events for **500+ participants**; used engagement metrics to optimize event scheduling and promotions, growing club membership by **40%**.
- Managed a cross-functional team of 10+ members across concurrent events; oversaw social media outreach and intercollegiate collaborations.

Content & Podcast Writer

Sep 2022 – Apr 2024

Farbe Club, SRM University AP

Amaravati, India

- Scripted podcasts and authored editorial content across platforms, boosting online engagement by **40%**.
- Collaborated with editorial teams on content campaigns, maintaining storytelling consistency and brand voice across digital channels.

Events & Writing Lead

Aug 2021 – May 2022

Student Council, SRM University AP

Amaravati, India

- Drafted event reports, communication briefs, and activity documentation; increased student participation by **30%** through structured promotion and coordinated outreach.
- Managed cross-team coordination ensuring accurate, timely documentation of all student council activities.